```python
import cv2
import numpy as np
from matplotlib import pyplot as plt
left_img  = cv2.imread(r"C:\Users\sreev\Downloads\left.png")
right_img = cv2.imread(r"C:\Users\sreev\Downloads\right.png")
if left_img is None or right_img is None:
    print("□ Error: Could not load images!")
else:
    print(f"✓ Left image size:  {left_img.shape}")
    print(f"✓ Right image size: {right_img.shape}")
left_gray  = cv2.cvtColor(left_img,  cv2.COLOR_BGR2GRAY)
right_gray = cv2.cvtColor(right_img, cv2.COLOR_BGR2GRAY)
orb = cv2.ORB_create(nfeatures=500)
kp1, des1 = orb.detectAndCompute(left_gray,  None)
kp2, des2 = orb.detectAndCompute(right_gray, None)
print(f"✓ Keypoints in Left Image:  {len(kp1)}")
print(f"✓ Keypoints in Right Image: {len(kp2)}")

bf      = cv2.BFMatcher(cv2.NORM_HAMMING, crossCheck=True)
matches = bf.match(des1, des2)
matches = sorted(matches, key=lambda x: x.distance)
print(f"✓ Total Matches Found: {len(matches)}")
print(f"✓ Best Match Distance: {matches[0].distance:.2f}")
good_matches = matches[:50]

matched_img = cv2.drawMatches(
    left_img,  kp1,
    right_img, kp2,
    good_matches, None,
    flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS
)
matched_rgb = cv2.cvtColor(matched_img, cv2.COLOR_BGR2RGB)
fig, axes = plt.subplots(1, 3, figsize=(20, 6))
fig.suptitle("AR Feature Matching Pipeline", fontsize=16,
fontweight='bold')
left_kp_img = cv2.drawKeypoints(left_img, kp1, None, color=(0, 255,
0))
axes[0].imshow(cv2.cvtColor(left_kp_img, cv2.COLOR_BGR2RGB))
axes[0].set_title(f"Left Image\n{len(kp1)} Keypoints", fontsize=12,
fontweight='bold')
axes[0].axis('off')
right_kp_img = cv2.drawKeypoints(right_img, kp2, None, color=(0, 255,
0))
axes[1].imshow(cv2.cvtColor(right_kp_img, cv2.COLOR_BGR2RGB))
axes[1].set_title(f"Right Image\n{len(kp2)} Keypoints", fontsize=12,
fontweight='bold')
axes[1].axis('off')
axes[2].imshow(matched_rgb)
axes[2].set_title(f"Feature Matches\nTop {len(good_matches)} Matches",
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fontsize=12, fontweight='bold')
axes[2].axis('off')
plt.tight_layout()
plt.show()

print("\n" + "="*50)
print("FEATURE MATCHING STATISTICS")
print("="*50)
print(f"Keypoints in Left Image  : {len(kp1)}")
print(f"Keypoints in Right Image : {len(kp2)}")
print(f"Total Matches            : {len(matches)}")
print(f"Good Matches (Top 50)    : {len(good_matches)}")
print(f"Best Match Distance      : {matches[0].distance:.2f}")
print(f"Worst Match Distance     : {matches[-1].distance:.2f}")
print("="*50)
print("\n✓ Feature Matching Complete!")

cv2.imwrite(r"C:\Users\sreev\Downloads\feature_matches.png",
            cv2.cvtColor(matched_rgb, cv2.COLOR_RGB2BGR))
print("✓ Result saved: Downloads/feature_matches.png")
```

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✓ Left image size:  (315, 408, 3)
✓ Right image size: (325, 405, 3)
✓ Keypoints in Left Image:  500
✓ Keypoints in Right Image: 500
✓ Total Matches Found: 163
✓ Best Match Distance: 17.00
```

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==================================================
FEATURE MATCHING STATISTICS
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Keypoints in Left Image  : 500
Keypoints in Right Image : 500
Total Matches            : 163
Good Matches (Top 50)    : 50
Best Match Distance      : 17.00
Worst Match Distance     : 85.00
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✓ Feature Matching Complete!
✓ Result saved: Downloads/feature_matches.png
```